

U-slot array antenna report at 110 GHz

For the first antenna topology (Antenna1), Fig. 1 shows a single element u-slot microstrip patch fed by a microstrip line that is designed and simulated. The patch dimension, slot dimension and the substrate thickness and permittivity is designed for optimum coupling.

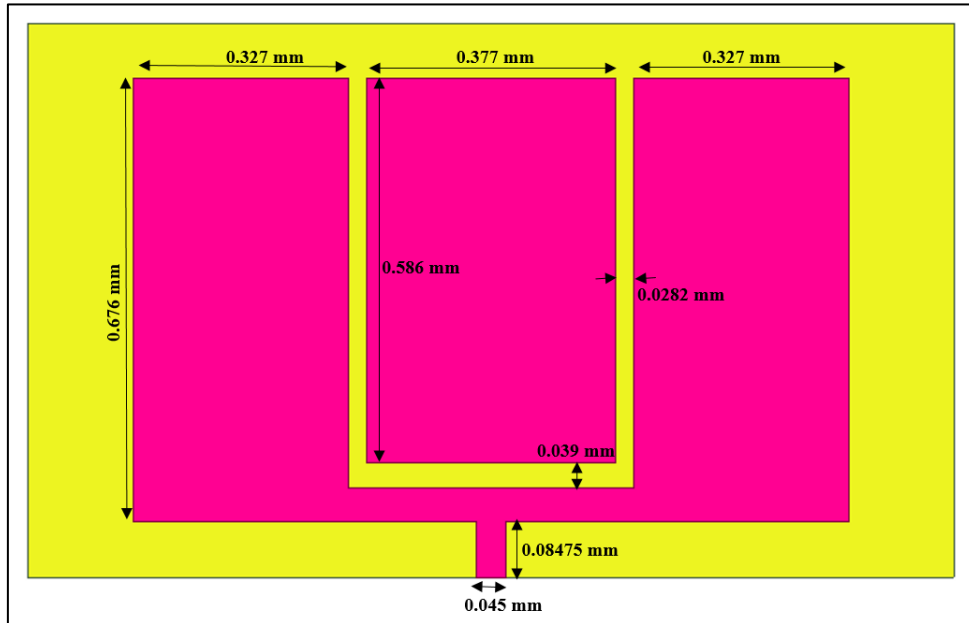


Fig.1. Single element u-slot microstrip patch antenna

The substrate, having a relative permittivity of 2.7 has dimensions of 0.85 mm in length, 1.45 mm in width, and 0.1973 mm in thickness. The U-slot is incorporated into the patch with a slot width of 0.0282 mm, while the horizontal arms of the U are 0.5866 mm each. The central vertical arm has a length of 0.377 mm, and the feed line width is 0.045 mm. The patch extends 0.676 mm vertically on the substrate. These dimensions are critical for the design and optimization of the antenna's performance.

For the second antenna topology (Antenna2), Fig. 1 shows a single element u-slot microstrip patch fed by a microstrip line that is designed and simulated on a substrate with relative permittivity of 10.2 and thickness of 0.197 mm

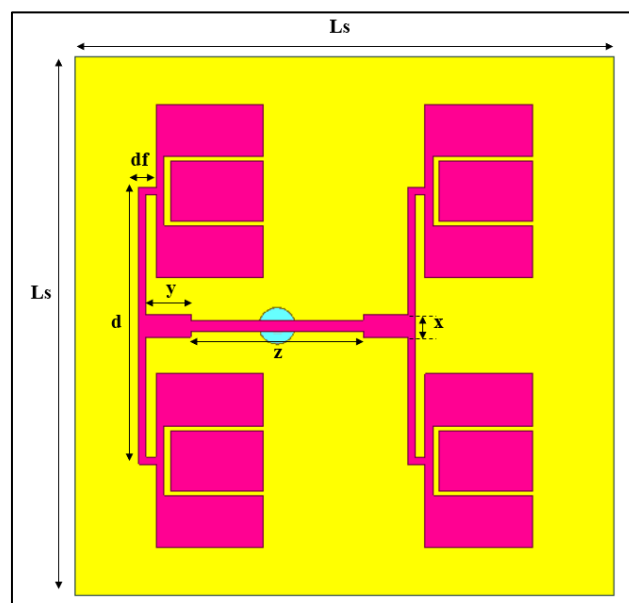


Fig.2. 2x2 array u-slot microstrip patch antenna

The figure depicts a U-slot microstrip patch array antenna that employs the same U-slot patch design as described previously. The array consists of four identical patches arranged symmetrically on a $4.2 \text{ mm} \times 4.2 \text{ mm}$ substrate. Each patch has dimensions consistent with the earlier description, including a U-slot width of 0.0282 mm , horizontal arm lengths of 0.327 mm , and a vertical arm length of 0.377 mm . The patches are spaced 1.736 mm apart vertically, while the feed network provides a separation of 1.08 mm horizontally between the rows. The feed line width is 0.14 mm , and the branch lines connecting to the patches have a width of 0.08475 mm . This configuration is designed to optimize the array's performance and enhance its radiation characteristics. The dimensions of the figure have been given in the following table.

The dimensions are tabulated in Table 1.

Table 1: Dimensions of the feed

Parameters	Description	Dimensions (mm)
Ls	Length of substrate	4.2
d	Horizontal separation between two patches	1.736
df	Length of planar extension	0.08475
x	Width of connector	0.14
y	Length of connector	0.282
z	Distance between two connectors	1.08

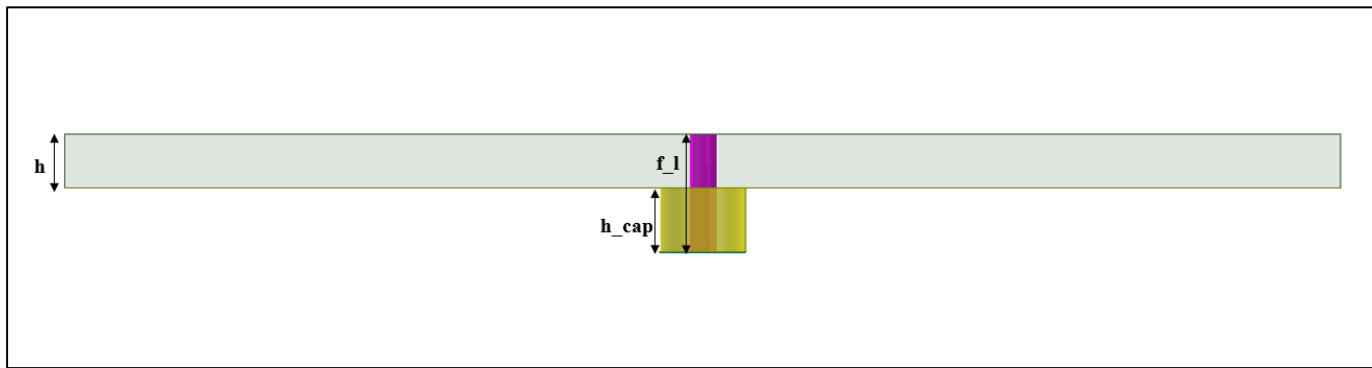


Fig.3. Side-view of the array antenna showing the feed line through the substrate.

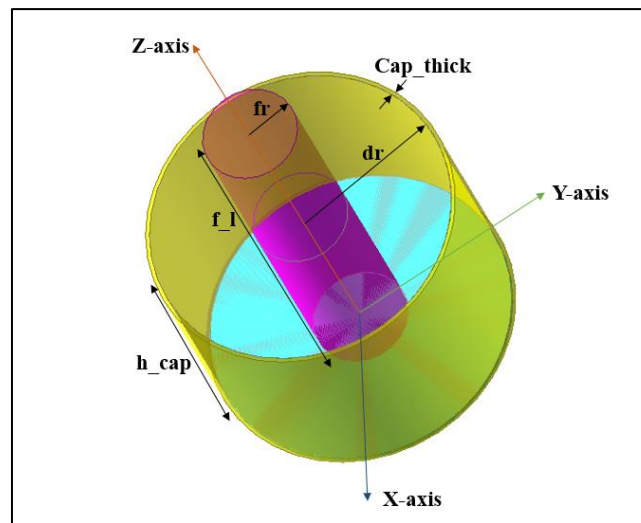


Fig.4. Trimetric view of the co-axial feed line.

The bottom-fed coaxial feed line for the U-slot microstrip patch array antenna consists of an inner conductor with a radius of 0.0355 mm . This conductor is surrounded by a dielectric material made of Teflon, which has a radius

of 0.112 mm. The coaxial feed line extends vertically from the ground plane, penetrating through the substrate to provide excitation to the patches. The Teflon dielectric ensures proper insulation and impedance matching, minimizing losses and enhancing the antenna's efficiency. This feed configuration is suitable for achieving uniform power distribution across the array and ensures robust performance at the desired frequency. The cap is terminated by a PEC layer with a thickness of 0.0028 mm.

The dimensions are tabulated in Table 1.

Table 1: Dimensions of the feed

Parameters	Description	Dimensions (mm)
h	Height of the substrate	0.1973
fr	Radius of feedline	0.0355
f_l	Length of feedline	0.31
h_cap	Height of feed cap	0.169
dr	Dielectric radius	0.112
Cap-thick	Thickness of the cap	0.0028

Results for Single element antenna:

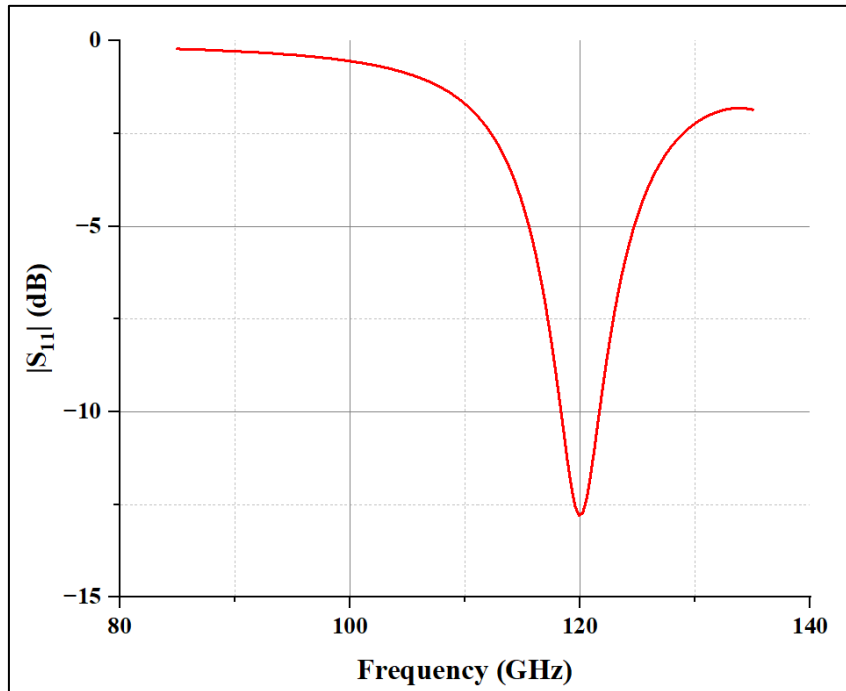


Fig. 5. Reflection coefficient vs frequency of a single element u-slot antenna.

The reflection coefficient curve for the single element u-slot patch antenna, as shown in fig. 5 gives us a maximum dip of **-12.6 dB at a resonant frequency of 120 GHz.**

The **percentage bandwidth** is calculated to be **2.7%.**

The Gain (dB) vs frequency curve as shown in fig. 6 gives us a **gain of 8 dB at the resonant frequency of 120 GHz.**

The VSWR curve as shown in fig. 7 shows that the single element u-slot antenna has a minimum **VSWR of 1.6** at 120 GHz.

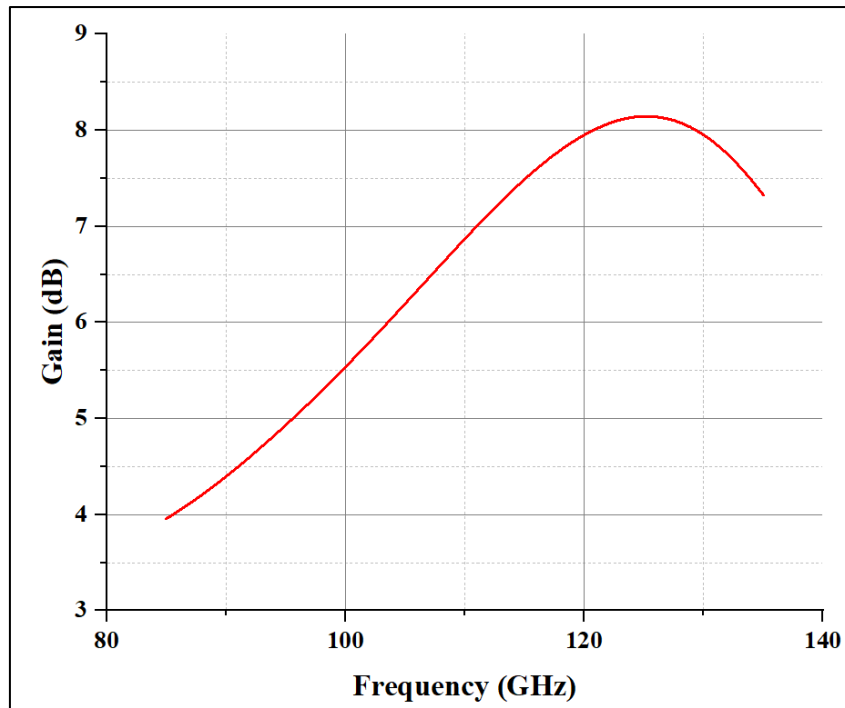


Fig. 6. Gain vs frequency of a single element u-slot antenna.

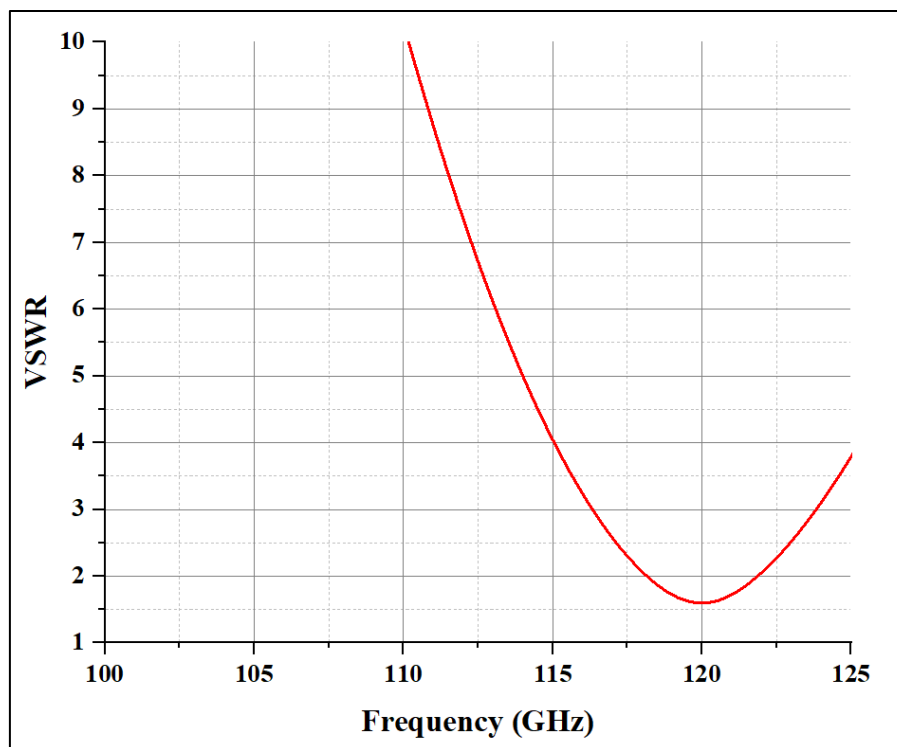
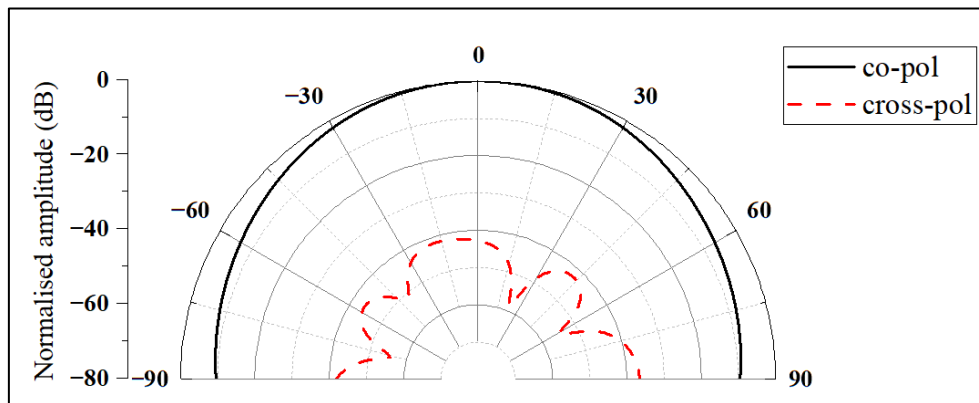
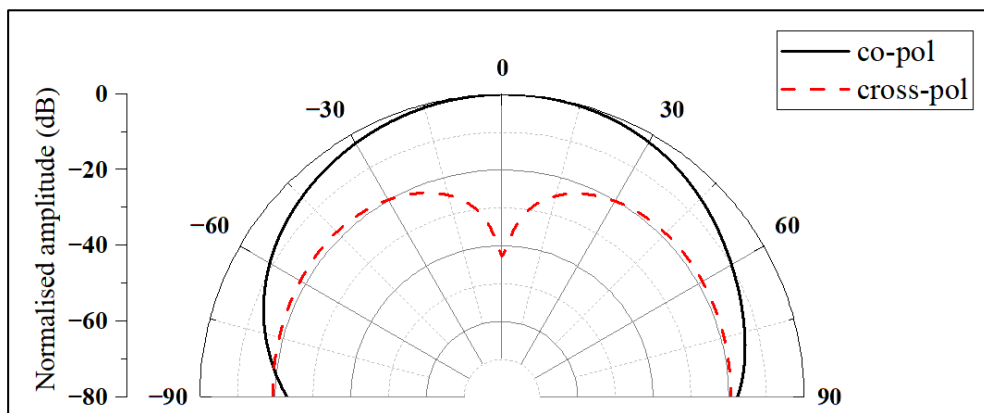


Fig. 7. VSWR vs frequency of a single element u-slot antenna.

The radiation patterns at the resonant frequency is shown in Fig. 8.



(a) $\phi = 0^\circ$



(b) $\phi = 90^\circ$

Fig. 8. Radiation pattern of a single element u-slot antenna at 120 GHz (a) $\phi = 0^\circ$ (b) $\phi = 90^\circ$

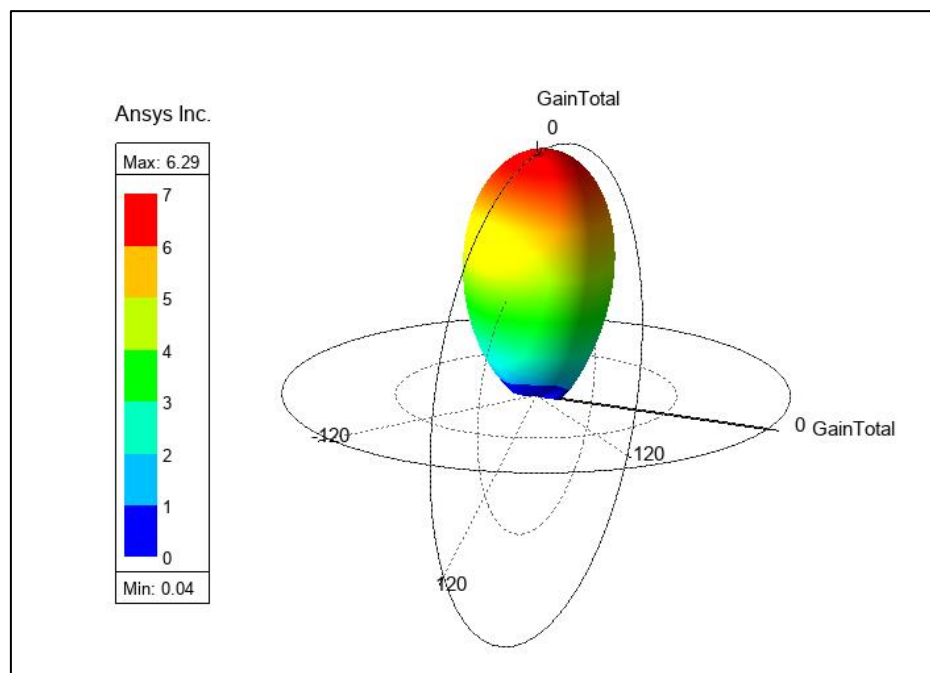


Fig. 9. 3-dimensional radiation pattern of 2x2 u-slot array antenna at 120 GHz.

The 3-dimensional radiation pattern in figure 9, shows a **maximum gain of 6.29 dB** of the 2x2 u-slot array antenna at the resonant frequency of 106.3 GHz.

Results for array antenna:

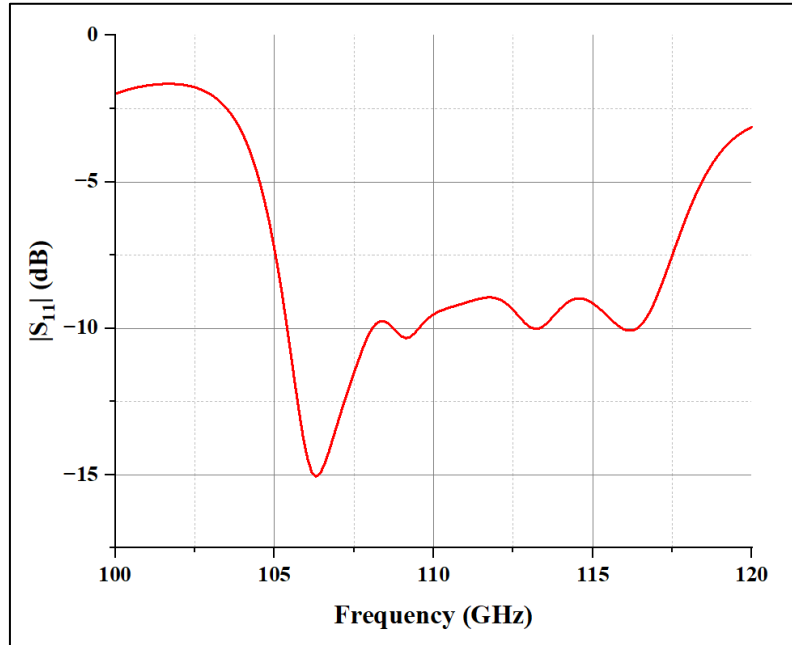


Fig. 10. Reflection coefficient vs frequency of 2x2 u-slot array antenna.

From fig.10, the resonant frequency is obtained at **106.3 GHz** with good coupling characteristics. Reflection coefficient of **-15 dB** is achieved.

The Gain (dB) vs frequency curve as shown in fig. 11 gives us a maximum **gain of 3.5 dB** at the resonant frequency of 106.3 GHz.

The VSWR curve as shown in fig. 12 shows that the single element u-slot antenna has a **minimum VSWR of 1.42 at 106.3 GHz**.

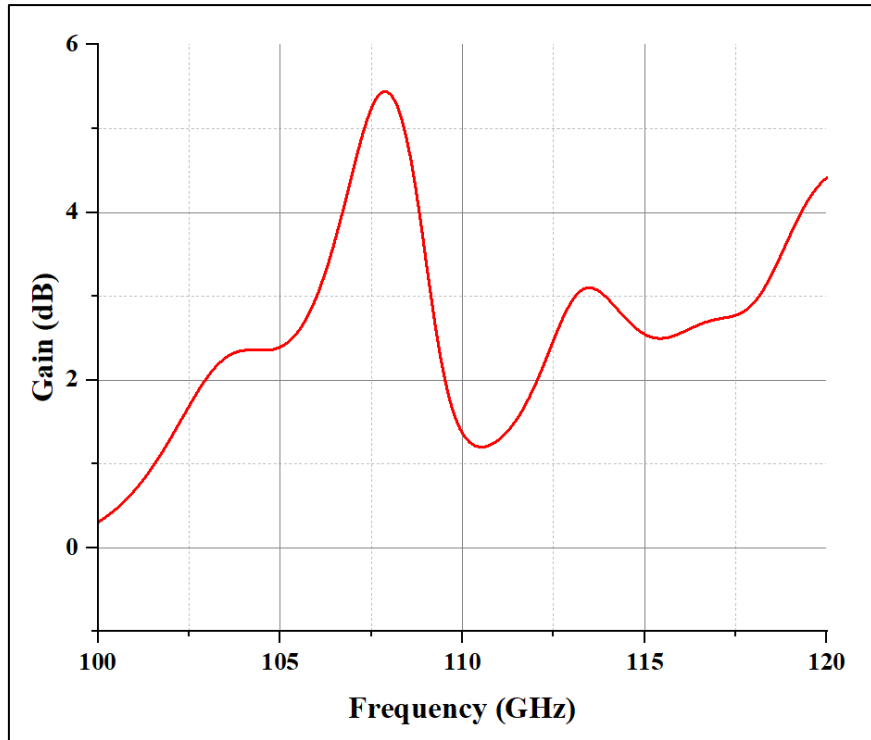


Fig. 11. Gain vs frequency of 2x2 u-slot array antenna.

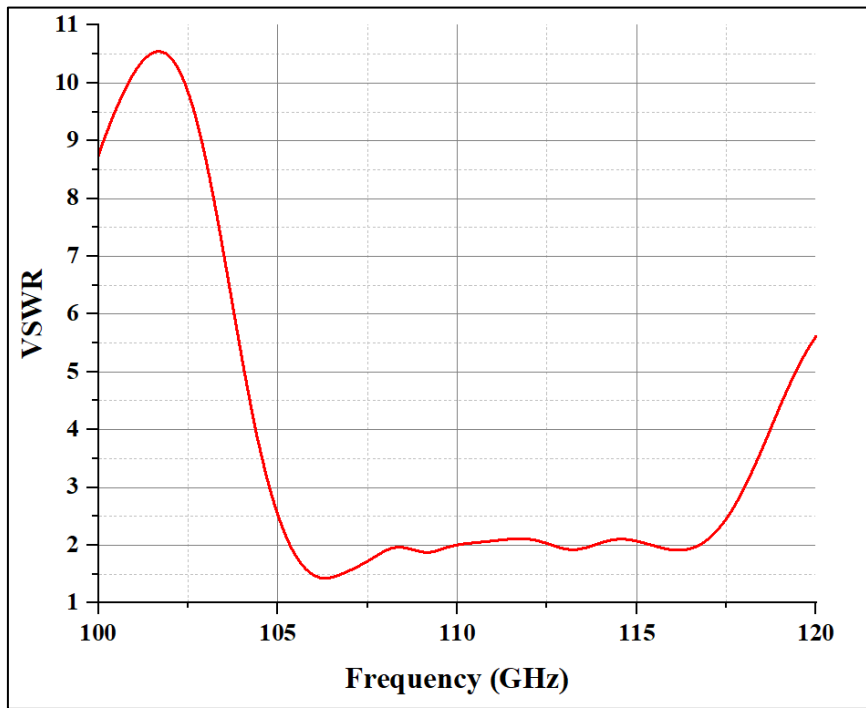
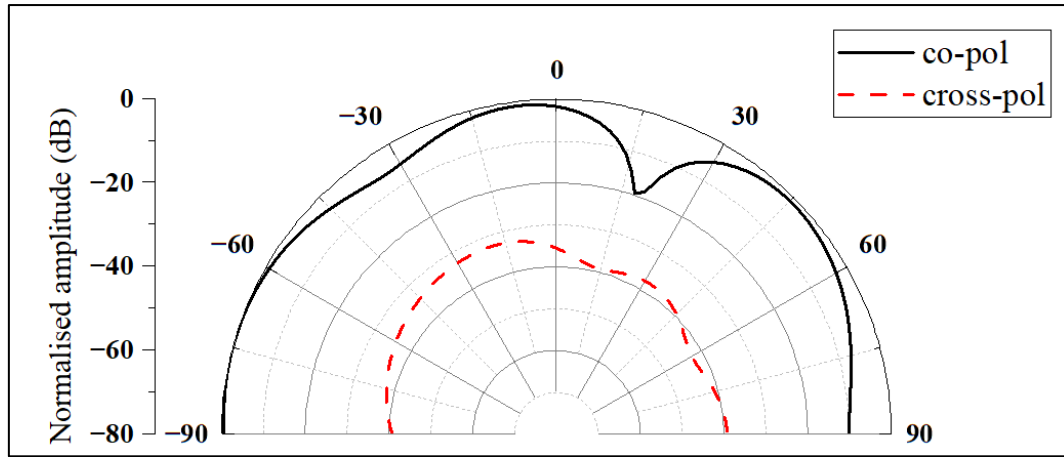
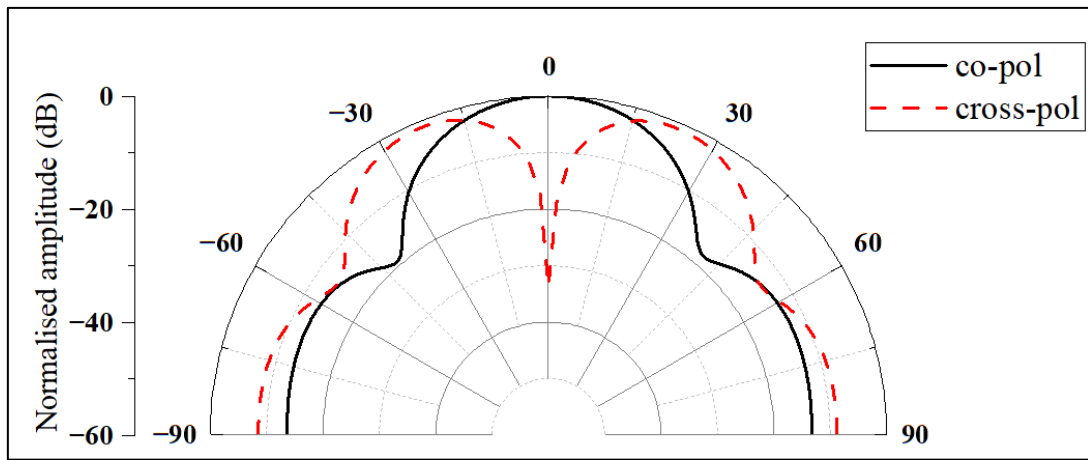


Fig. 12. VSWR vs frequency of 2x2 u-slot array antenna.

The radiation patterns at the resonant frequency is shown in Fig. 13.



(a) $\phi = 0^\circ$



(b) $\phi = 90^\circ$

Fig. 13. Radiation pattern of 2x2 u-slot array antenna at 106.3 GHz (a) $\phi = 0^\circ$ (b) $\phi = 90^\circ$

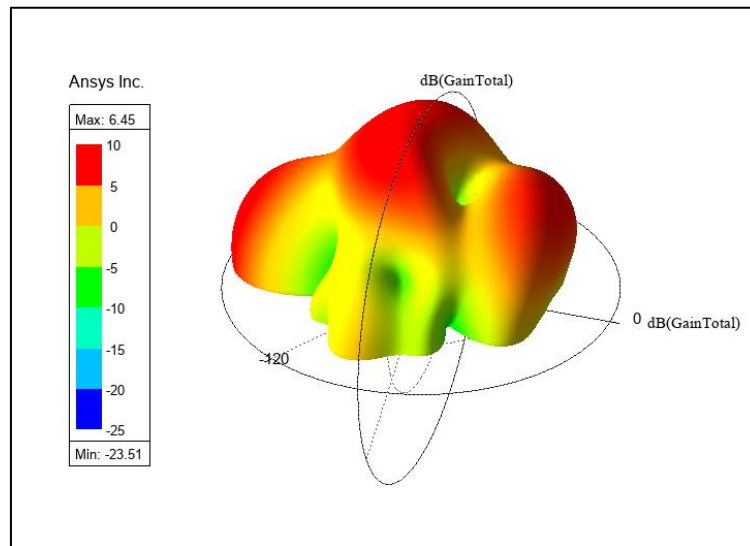


Fig. 14. 3-dimensional radiation pattern of 2x2 u-slot array antenna at 106.3 GHz.

The 3-dimensional radiation pattern in figure 14, shows a **maximum gain of 6.45 dB** of the 2x2 u-slot array antenna at the resonant frequency of 106.3 GHz.

